

4.1 Graphical Models

Graphical Models (GMs) are depictions of independence/dependence relationships for distributions. Each form of GM is a particular union of graph and probability constructs and details the form of independence assumptions represented. GMs are useful since they provide a framework for studying a wide class of probabilistic models and associated algorithms. In particular they help to clarify modelling assumptions and provide a unified framework under which inference algorithms in different communities can be related.

It needs to be emphasised that all forms of GM have a limited ability to graphically express conditional (in)dependence statements[265]. As we've seen, Belief Networks are useful for modelling ancestral conditional independence. In this chapter we'll introduce other types of GM that are more suited to representing different assumptions. Markov Networks, for example, are particularly suited to modelling marginal dependence and conditional independence. Here we'll focus on Markov Networks, Chain Graphs (which marry Belief and Markov networks) and Factor Graphs. There are many more inhabitants of the zoo of Graphical Models, see [70, 293].

The general viewpoint we adopt is to describe the problem environment using a probabilistic model, after which reasoning corresponds to performing probabilistic inference. This is therefore a two part process :

Modelling After identifying all potentially relevant variables of a problem environment, our task is to describe how these variables can interact. This is achieved using structural assumptions as to the form of the joint probability distribution of all the variables, typically corresponding to assumptions of independence of variables. Each class of graphical model corresponds to a factorisation property of the joint distribution.

Inference Once the basic assumptions as to how variables interact with each other is formed (*i.e.* the probabilistic model is constructed) all questions of interest are answered by performing inference on the distribution. This can be a computationally non-trivial step so that coupling GMs with accurate inference algorithms is central to successful graphical modelling.

Whilst not a strict separation, GMs tend to fall into two broad classes – those useful in modelling, and those useful in representing inference algorithms. For modelling, Belief Networks, Markov Networks, Chain Graphs and Influence Diagrams are some of the most popular. For inference one typically 'compiles' a model into a suitable GM for which an algorithm can be readily applied. Such inference GMs include Factor Graphs, Junction Trees and Region Graphs.

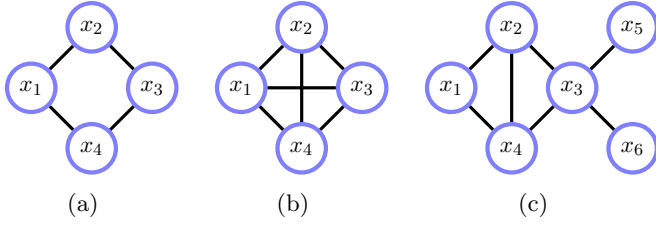


Figure 4.1: (a): $p_a = \phi(x_1, x_2)\phi(x_2, x_3)\phi(x_3, x_4)\phi(x_4, x_1)/Z_a$. (b): $p_b = \phi(x_1, x_2, x_3, x_4)/Z_b$. (c): $p_c = \phi(x_1, x_2, x_4)\phi(x_2, x_3, x_4)\phi(x_3, x_5)\phi(x_3, x_6)/Z_c$.

4.2 Markov Networks

Belief Networks correspond to a special kind of factorisation of the joint probability distribution in which each of the factors is itself a distribution. An alternative factorisation is, for example

$$p(a, b, c) = \frac{1}{Z} \phi(a, b) \phi(b, c) \quad (4.2.1)$$

where $\phi(a, b)$ and $\phi(b, c)$ are *potentials* and Z is a constant which ensures normalisation, called the *partition function*

$$Z = \sum_{a, b, c} \phi(a, b) \phi(b, c) \quad (4.2.2)$$

We will typically use the convention that the ordering of the variables in the potential is not relevant (as for a distribution) – the joint variables simply index an element of the potential table. Markov Networks are defined as products of non-negative functions defined on maximal cliques of an undirected graph – see fig(4.1).

Definition 24 (Potential). A potential $\phi(x)$ is a non-negative function of the variable x , $\phi(x) \geq 0$. A joint potential $\phi(x_1, \dots, x_n)$ is a non-negative function of the set of variables. A distribution is a special case of a potential satisfying normalisation, $\sum_x \phi(x) = 1$. This holds similarly for continuous variables, with summation replaced by integration.

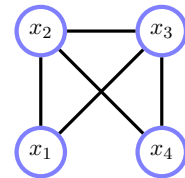
Definition 25 (Markov Network). For a set of variables $\mathcal{X} = \{x_1, \dots, x_n\}$ a Markov network is defined as a product of potentials on subsets of the variables $\mathcal{X}_c \subseteq \mathcal{X}$:

$$p(x_1, \dots, x_n) = \frac{1}{Z} \prod_{c=1}^C \phi_c(\mathcal{X}_c) \quad (4.2.3)$$

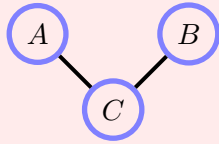
Graphically this is represented by an undirected graph G with $\mathcal{X}_c, c = 1, \dots, C$ being the maximal cliques of G . The constant Z ensures the distribution is normalised. The graph is said to satisfy the factorisation property. In the special case that the graph contains cliques of only size 2, the distribution is called a *pairwise Markov Network*, with potentials defined on each link between two variables.

For the case in which clique potentials are strictly positive, this is called a *Gibbs distribution*.

Remark 4 (*Pairwise Markov network*). Whilst a Markov network is formally defined on maximal cliques, in practice authors often use the term to refer to non-maximal cliques. For example, in the graph on the right, the maximal cliques are x_1, x_2, x_3 and x_2, x_3, x_4 , so that the graph describes a distribution $p(x_1, x_2, x_3, x_4) = \phi(x_1, x_2, x_3)\phi(x_2, x_3, x_4)/Z$. In a pairwise network though the potentials are assumed to be over two-cliques, giving $p(x_1, x_2, x_3, x_4) = \phi(x_1, x_2)\phi(x_1, x_3)\phi(x_2, x_3)\phi(x_2, x_4)\phi(x_3, x_4)/Z$.



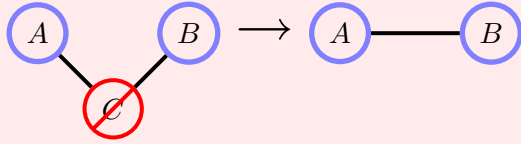
Definition 26 (Properties of Markov Networks).



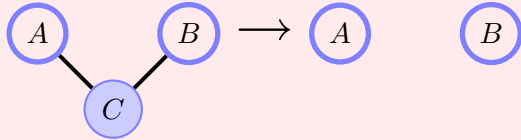
$$p(A, B, C) = \phi_{AC}(A, C)\phi_{BC}(B, C)/Z \quad (4.2.4)$$

A and B are unconditionally dependent : $p(A, B) \neq p(A)p(B)$.

A and B are conditionally independent on C : $p(A, B|C) = p(A|C)p(B|C)$.



Marginalising over C makes A and B (graphically) dependent.



Conditioning on C makes A and B independent.

4.2.1 Markov properties

We here state some of the most useful results. The reader is referred to [168] for proofs and more detailed discussion. Consider the Markov Network in fig(4.2a). Here we use the shorthand $p(1) \equiv p(x_1)$, $\phi(1, 2, 3) \equiv \phi(x_1, x_2, x_3)$ etc. We will use this undirected graph to demonstrate conditional independence properties.

Local Markov property

Definition 27 (Local Markov Property).

$$p(x|\mathcal{X} \setminus x) = p(x|\text{ne}(x)) \quad (4.2.5)$$

When conditioned on its neighbours, x is independent of the remaining variables of the graph.

The conditional distribution $p(4|1, 2, 3, 5, 6, 7)$ is

$$p(4|1, 2, 3, 5, 6, 7) = \frac{\phi(1, 2, 3)\phi(2, 3, 4)\phi(4, 5, 6)\phi(5, 6, 7)}{\sum_4 \phi(1, 2, 3)\phi(2, 3, 4)\phi(4, 5, 6)\phi(5, 6, 7)} = \frac{\phi(2, 3, 4)\phi(4, 5, 6)}{\sum_4 \phi(2, 3, 4)\phi(4, 5, 6)} = p(4|2, 3, 5, 6) \quad (4.2.6)$$

The last line above follows since the variable x_4 only appears in the cliques that border x_4 . The generalisation of the above example is clear: a MN with positive clique potentials ϕ , defined with respect to an undirected graph G entails¹ $p(x_i|x_{\setminus i}) = p(x_i|\text{ne}(x_i))$.

Pairwise Markov property

Definition 28 (Pairwise Markov Property). For any non-adjacent vertices x and y

$$x \perp\!\!\!\perp y | \mathcal{X} \setminus \{x, y\} \quad (4.2.7)$$

¹The notation $x_{\setminus i}$ is shorthand for the set of all variables $\mathcal{X}$ excluding variable x_i , namely $\mathcal{X} \setminus x_i$ in set notation.

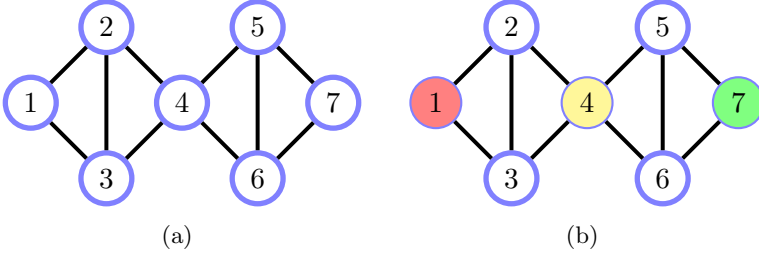


Figure 4.2: (a): $\phi(1, 2, 3)\phi(2, 3, 4)\phi(4, 5, 6)\phi(5, 6, 7)$. (b): By the global Markov property, since every path from 1 to 7 passes through 4, then $1 \perp\!\!\!\perp 7 | 4$.

$$p(1, 4 | 2, 3, 5, 6, 7) = \frac{\phi(1, 2, 3)\phi(2, 3, 4)\phi(4, 5, 6)\phi(5, 6, 7)}{\sum_{1,4} \phi(1, 2, 3)\phi(2, 3, 4)\phi(4, 5, 6)\phi(5, 6, 7)} = \frac{\phi(1, 2, 3)\phi(2, 3, 4)\phi(4, 5, 6)}{\sum_{1,4} \phi(1, 2, 3)\phi(2, 3, 4)\phi(4, 5, 6)} \quad (4.2.8)$$

$$= p(1 | 2, 3, 4, 5, 6, 7) p(4 | 1, 2, 3, 5, 6, 7) \quad (4.2.9)$$

where the last line follows since for fixed 2, 3, 5, 6, 7, the function $\phi(1, 2, 3)\phi(2, 3, 4)\phi(4, 5, 6)$ is a product of a function on 1 and a function on 4, implying independence.

Global Markov property

Definition 29 (Separation). A subset $\mathcal{S}$ separates a subset $\mathcal{A}$ from a subset $\mathcal{B}$ if every path from any member of $\mathcal{A}$ to any member of $\mathcal{B}$ passes through $\mathcal{S}$.

Definition 30 (Global Markov Property). For a disjoint subset of variables, $(\mathcal{A}, \mathcal{B}, \mathcal{S})$ where $\mathcal{S}$ separates $\mathcal{A}$ from $\mathcal{B}$ in G , then $\mathcal{A} \perp\!\!\!\perp \mathcal{B} | \mathcal{S}$.

$$p(1, 7 | 4) \propto \sum_{2,3,5,6} p(1, 2, 3, 4, 5, 6, 7) \quad (4.2.10)$$

$$= \sum_{2,3,5,6} \phi(1, 2, 3)\phi(2, 3, 4)\phi(4, 5, 6)\phi(5, 6, 7) \quad (4.2.11)$$

$$= \left\{ \sum_{2,3} \phi(1, 2, 3)\phi(2, 3, 4) \right\} \left\{ \sum_{5,6} \phi(4, 5, 6)\phi(5, 6, 7) \right\} \quad (4.2.12)$$

This implies that $p(1, 7 | 4) = p(1 | 4)p(7 | 4)$.

Example 16 (*Boltzmann machine*). A Boltzmann machine is a MN on binary variables $\text{dom}(x_i) = \{0, 1\}$ of the form

$$p(\mathbf{x}) = \frac{1}{Z(\mathbf{w}, \mathbf{b})} e^{\sum_{i < j} w_{ij} x_i x_j + \sum_i b_i x_i} \quad (4.2.13)$$

where the interactions w_{ij} are the ‘weights’ and the b_i the biases. This model has been studied in the machine learning community as a basic model of distributed memory and computation[2]. The graphical model of the BM is an undirected graph with a link between nodes i and j for $w_{ij} \neq 0$. Consequently, for all but specially constrained $\mathbf{W}$, the graph is multiply-connected and inference will be typically intractable.

4.2.2 Gibbs networks

For simplicity we assume that the potentials are strictly positive in which case MNs are also termed *Gibbs Networks*. In this case, a GN satisfies the following independence relations:

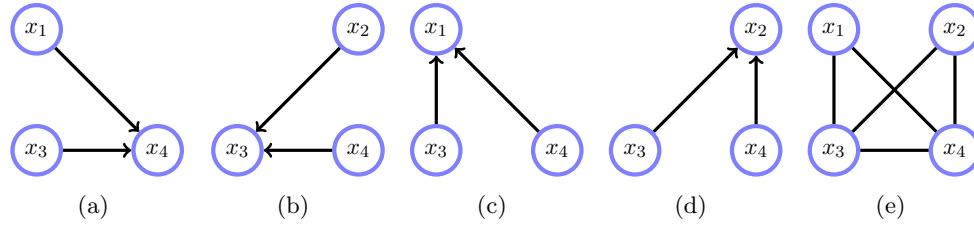


Figure 4.3: **(a-d)**: Local distributions. **(e)**: The Markov network consistent with the local distributions. If the local distributions are positive, by the Hammersley-Clifford theorem, the only joint distribution that can be consistent with the local distributions must be a Gibbs distribution with structure given by (e).

4.2.3 Markov random fields

Definition 31 (Markov Random Field). A MRF is defined by a set of distributions $p(x_i | \text{ne}(x_i))$ where $i \in \{1, \dots, n\}$ indexes the distributions and $\text{ne}(x_i)$ are the neighbours of variable x_i , namely that subset of the variables $x_1, \dots, x_n$ that the distribution of variable x_i depends on. The term Markov indicates that this is a proper subset of the variables.

A distribution is an MRF with respect to an undirected graph G if

$$p(x_i | x_{\setminus i}) = p(x_i | \text{ne}(x_i)) \quad (4.2.14)$$

where $\text{ne}(x_i)$ are the neighbouring variables of variable x_i , according to the undirected graph G .

Hammersley Clifford Theorem

The Hammersley-Clifford theorem helps resolve questions as to when a set of positive local distributions $p(x_i | \text{ne}(x_i))$ could ever form a consistent joint distribution $p(x_1, \dots, x_n)$. Local distributions $p(x_i | \text{ne}(x_i))$ can form a consistent joint distribution if and only if $p(x_1, \dots, x_n)$ factorises according to

$$p(x_1, \dots, x_n) = \frac{1}{Z} \exp \left(- \sum_c V_c(\mathcal{X}_c) \right) \quad (4.2.15)$$

where the sum is over all cliques and $V_c(\mathcal{X}_c)$ is a real function defined over the variables in the clique indexed by c . Equation (4.2.15) is equivalent to $\prod_c \phi(\mathcal{X}_c)$, namely a MN on positive clique potentials.

The graph over which the cliques are defined is an undirected graph with a link between x_i and x_j if

$$p(x_i | x_{\setminus i}) \neq p(x_i | x_{\setminus (i,j)}) \quad (4.2.16)$$

That is, if x_j has an effect on the conditional distribution of x_i , then add an undirected link between x_i and x_j . This is then repeated over all the variables x_i [35, 203], see fig(4.3). Note that the HC theorem does not mean that given a set of conditional distributions, we can always form a consistent joint distribution from them – rather it states what the functional form of a joint distribution for the conditionals to be consistent with the joint, see exercise(45).

4.2.4 Conditional independence using Markov networks

For $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ each being collections of variables, in section(3.3.3) we discussed an algorithm to determine $\mathcal{X} \perp\!\!\!\perp \mathcal{Y} | \mathcal{Z}$. An alternative and more general method (since it handles directed and undirected graphs) uses the following steps: (see [74, 169])

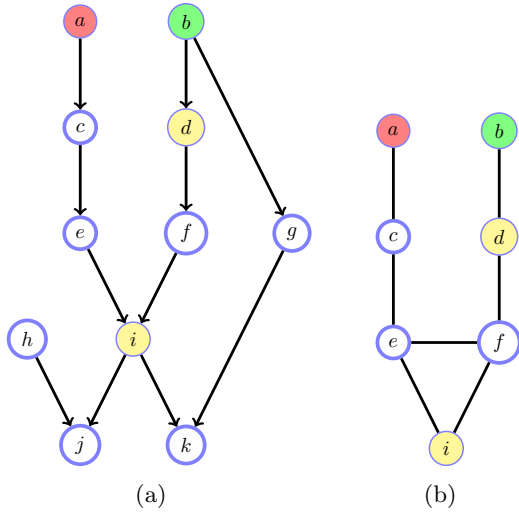


Figure 4.4: **(a)**: Belief Network for which we are interested in checking conditional independence $a \perp\!\!\!\perp b \mid \{d, i\}$. **(b)**: Ancestral moralised graph for $a \perp\!\!\!\perp b \mid \{d, i\}$. Every path from a red to green node passes through a yellow node, so a and b are independent given d, i . Alternatively, if we consider $a \perp\!\!\!\perp b \mid i$, the variable d is uncoloured, and we can travel from the red to the green without encountering a yellow node (using the $e - f$ path). In this case a is dependent on b , conditioned on i .

Ancestral Graph Remove from the DAG any node which is neither in $\mathcal{X} \cup \mathcal{Y} \cup \mathcal{Z}$ nor an ancestor of a node in this set, together with any edges in or out of such nodes.

Moralisation Add a line between any two remaining nodes which have a common child, but are not already connected by an arrow. Then remove remaining arrowheads.

Separation In the undirected graph so constructed, look for a path which joins a node in $\mathcal{X}$ to one in $\mathcal{Y}$ but does not intersect $\mathcal{Z}$. If there is no such path deduce that $\mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mid \mathcal{Z}$.

For Markov Networks only the final separation criterion needs to be applied. See fig(4.4) for an example.

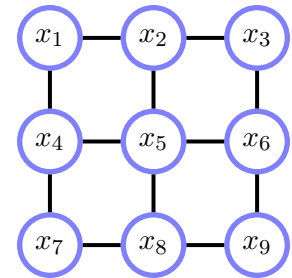
4.2.5 Lattice Models

Undirected models have a long history in different branches of science, especially statistical mechanics on lattices and more recently as models in visual processing in which the models encourage neighbouring variables to be in the same states[35, 36, 106].

Consider a model in which our desire is that states of the binary valued variables $x_1, \dots, x_9$, arranged on a lattice (right) should prefer their neighbouring variables to be in the same state

$$p(x_1, \dots, x_9) = \frac{1}{Z} \prod_{i \sim j} \phi_{ij}(x_i, x_j) \quad (4.2.17)$$

where $i \sim j$ denotes the set of indices where i and j are neighbours in the undirected graph.



The Ising model

A set of potentials for equation (4.2.17) that encourages neighbouring variables to have the same state is

$$\phi_{ij}(x_i, x_j) = e^{-\frac{1}{2T}(x_i - x_j)^2} \quad (4.2.18)$$

This corresponds to a well-known model of the physics of magnetic systems, called the *Ising model* which consists of ‘mini-magnets’ which prefer to be aligned in the same state, depending on the temperature

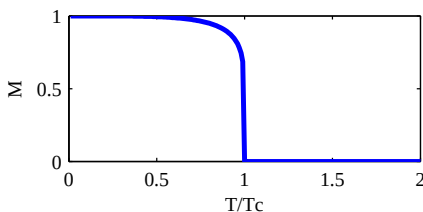


Figure 4.5: Onsager magnetisation. As the temperature T decreases towards the critical temperature T_c a phase transition occurs in which a large fraction of the variables become aligned in the same state.

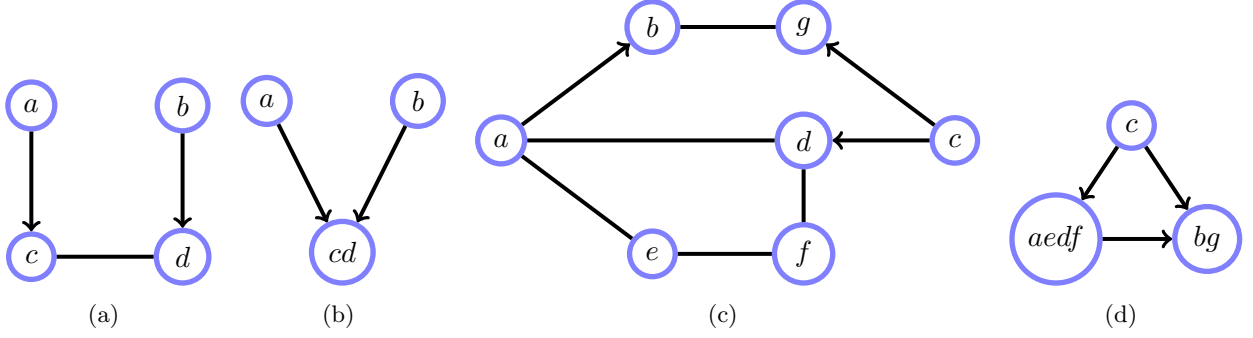


Figure 4.6: Chain graphs. The chain components are identified by deleting the directed edges and identifying the remaining connected components. (a): Chain components are $(a), (b), (c, d)$, which can be written as a BN on the cluster variables in (b). (c): Chain components are $(a, e, d, f), (b, g), (c)$, which has the cluster BN representation (d). (From [168])

T . For high T the variables behave independently so that no global magnetisation appears. For low T , there is a strong preference for neighbouring mini-magnets to become aligned, generating a strong macro-magnet. Remarkably, one can show that, in a very large two-dimensional lattice, below the so-called Curie temperature, $T_c \approx 2.269$ (for ± 1 variables), the system admits a phase change in that a large fraction of the variables become aligned – above T_c , on average, the variables are unaligned. This is depicted in fig(4.5) where $M = |\sum_{i=1}^N x_i|/N$ is the average alignment of the variables. That this phase change happens for non-zero temperature has driven considerable research in this and related areas[40]. Global coherence effects such as this that arise from weak local constraints are present in systems that admit *emergent behaviour*. Similar local constraints are popular in image restoration algorithms to clean up noise, under the assumption that noise will not show any local spatial coherence, whilst ‘signal’ will. An example is given in section(28.8) where we discuss algorithms for inference under special constraints on the MRF.

4.3 Chain Graphical Models

Definition 32 (Chain Component). The chain components of a graph G are obtained by :

1. Forming a graph G' with directed edges removed from G .
2. Then each connected component in G' constitutes a chain component.

Chain Graphs (CGs) contain both directed and undirected links. To develop the intuition, consider fig(4.6a). The only terms that we can unambiguously specify from this depiction are $p(a)$ and $p(b)$ since there is no mixed interaction of directed and undirected edges at the a and b vertices. By probability, therefore, we must have

$$p(a, b, c, d) = p(a)p(b)p(c, d|a, b) \quad (4.3.1)$$

Looking at the graph, we might expect the interpretation to be

$$p(c, d|a, b) = \phi(c, d)p(c|a)p(d|b) \quad (4.3.2)$$

However, to ensure normalisation, and also to retain generality, we interpret this chain component as

$$p(c, d|a, b) = \phi(c, d)p(c|a)p(d|b)\phi(a, b), \text{ with } \phi(a, b) \equiv 1 / \sum_{c, d} \phi(c, d)p(c|a)p(d|b) \quad (4.3.3)$$

This leads to the interpretation of a CG as a DAG over the chain components. Each chain component represents a distribution over the variables of the component, conditioned on the parental components. The conditional distribution is itself a product over the cliques of the undirected component and moralised parental components, including also a factor to ensure normalisation over the chain component.

Definition 33 (Chain Graph distribution). The distribution associated with a chain graph G is found by first identifying the chain components, τ . Then

$$p(x) = \prod_{\tau} p(\mathcal{X}_{\tau} | \text{pa}(\mathcal{X}_{\tau})) \quad (4.3.4)$$

and

$$p(\mathcal{X}_{\tau} | \text{pa}(\mathcal{X}_{\tau})) \propto \prod_{c \in \mathcal{C}_{\tau}} \phi(\mathcal{X}_{\mathcal{C}_{\tau}}) \quad (4.3.5)$$

where $\mathcal{C}_{\tau}$ denotes the union of the cliques in component τ together with the moralised parental components of τ , with ϕ being the associated functions defined on each clique. The proportionality factor is determined implicitly by the constraint that the distribution sums to 1.

BNs are CGs in which the connected components are singletons. MNs are CGs in which the chain components are simply the connected components of the undirected graph.

CGs can be useful since they are more expressive of CI statements than either Belief Networks or Markov Networks alone. The reader is referred to [168] and [99] for further details.

Example 17 (Chain Graphs are more expressive than Belief or Markov Networks). Consider the chain graph in fig(4.7a), which has chain component decomposition

$$p(a, b, c, d, e, f) = p(a)p(b)p(c, d, e, f | a, b) \quad (4.3.6)$$

where

$$p(c, d, e, f | a, b) = \phi(a, c)\phi(c, e)\phi(e, f)\phi(d, f)\phi(d, b)\phi(a, b) \quad (4.3.7)$$

with the normalisation requirement

$$\phi(a, b) \equiv 1 / \sum_{c, d, e, f} \phi(a, c)\phi(c, e)\phi(e, f)\phi(d, f)\phi(d, b) \quad (4.3.8)$$

The marginal $p(c, d, e, f)$ is given by

$$\phi(c, e)\phi(e, f)\phi(d, f) \underbrace{\sum_{a, b} \phi(a, b)p(a)p(b)\phi(a, c)\phi(d, b)}_{\phi(c, d)} \quad (4.3.9)$$

Since the marginal distribution of $p(c, d, e, f)$ is an undirected 4-cycle, no DAG can express the CI statements contained in the marginal $p(c, d, e, f)$. Similarly no undirected distribution on the same skeleton as fig(4.7a) could express that a and b are independent (unconditionally), *i.e.* $p(a, b) = p(a)p(b)$.

4.4 Expressiveness of Graphical Models

It is clear that directed distributions can be represented as undirected distributions since one can associate each (normalised) factor in a directed distribution with a potential. For example, the distribution $p(a|b)p(b|c)p(c)$ can be factored as $\phi(a, b)\phi(b, c)$, where $\phi(a, b) = p(a|b)$ and $\phi(b, c) = p(b|c)p(c)$, with $Z = 1$. Hence every Belief Network can be represented as some MN by simple identification of the factors in the distributions. However, in general, the associated undirected graph (which corresponds to the

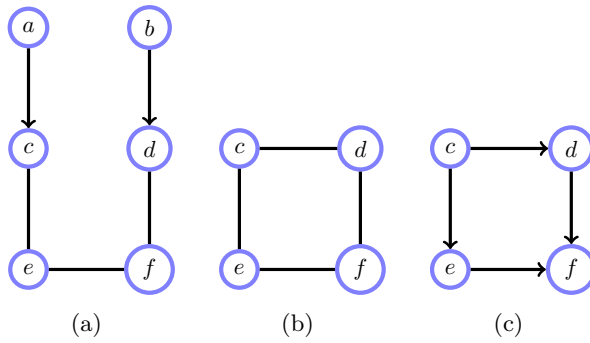


Figure 4.7: The CG (a) expresses $a \perp\!\!\!\perp b \mid \emptyset$ and $d \perp\!\!\!\perp e \mid (c, f)$. No directed graph could express both these conditions since the marginal distribution $p(c, d, e, f)$ is an undirected four cycle, (b). Any DAG on a 4 cycle must contain a collider, as in (c) and therefore express a different set of CI statements than (b). Similarly, no connected Markov network can express unconditional independence and hence (a) expresses CI statements that no Belief Network or Markov Network alone can express.

moralised directed graph) will contain additional links and independence information can be lost. For example, the MN of $p(c|a, b)p(a)p(b)$ if a single clique $\phi(a, b, c)$ from which one cannot graphically infer that $a \perp\!\!\!\perp b \mid \emptyset$.

The converse question is whether every undirected model can be represented by a BN with a readily derived link structure? Consider the example in fig(4.8). In this case, there is no directed model with the same link structure that can express the (in)dependencies in the undirected graph. Naturally, *every* probability distribution can be represented by some BN though it may not necessarily have a simple structure and be simply a ‘fully connected’ cascade style graph. In this sense the DAG cannot graphically represent the independence/dependence relations true in the distribution.

Definition 34 (Independence Maps). A graph is an *independence map* (I-map) of a given distribution P if every conditional independence statement that one can derive from the graph G is true in the distribution P . That is

$$\mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mid \mathcal{Z}_G \Rightarrow \mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mid \mathcal{Z}_P \quad (4.4.1)$$

for all disjoint sets $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$.

Similarly, a graph is a *dependence map* (D-map) of a given distribution P if every conditional independence statement that one can derive from P is true in the graph G . That is

$$\mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mid \mathcal{Z}_G \Leftarrow \mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mid \mathcal{Z}_P \quad (4.4.2)$$

for all disjoint sets $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$.

A graph G which is both an I-map and a D-map for P is called a *perfect map* and

$$\mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mid \mathcal{Z}_G \Leftrightarrow \mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mid \mathcal{Z}_P \quad (4.4.3)$$

for all disjoint sets $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$. In this case, the set of all conditional independence and dependence statements expressible in the graph G are consistent with P and *vice versa*.

Due to Inverse Modus Ponens, example(5), a dependence map is equivalent to

$$\mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mid \mathcal{Z}_G \Rightarrow \mathcal{X} \perp\!\!\!\perp \mathcal{Y} \mid \mathcal{Z}_P \quad (4.4.4)$$

although this is less useful since standard graphical model representations cannot express dependence.

Note that the above definitions are not dependent on the graph being directed or undirected. Indeed, some distributions may have a perfect directed map, but no perfect undirected map. For example

$$p(x, y, z) = p(z|x, y)p(x)p(y) \quad (4.4.5)$$

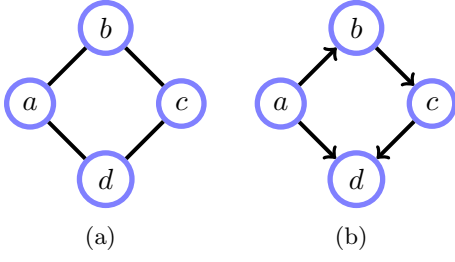


Figure 4.8: **(a)**: An undirected model for which we wish to find a directed equivalent. **(b)**: Every DAG with the same structure as the undirected model must have a situation where two arrows will point to a node, such as node d . Summing over the states of variable d will leave a DAG on the variables a, b, c with no link between a and c . This cannot represent the undirected model since when one marginalizes over d in the undirected this adds a link between a and c .

has a directed perfect map $x \rightarrow z \leftarrow y$ (assuming that $p(z|x, y) \neq \phi_x(x)\phi_y(y)$), but no perfect undirected map.

Example 18. Consider the distribution defined on variables t_1, t_2, y_1, y_2 [232]:

$$p(t_1, t_2, y_1, y_2) = p(t_1)p(t_2) \sum_h p(y_1|t_1, h)p(y_2|t_2, h)p(h) \quad (4.4.6)$$

The BN

$$p(y_2|y_1, t_2)p(y_1|t_1)p(t_1)p(t_2) \quad (4.4.7)$$

is an I-MAP for distribution (4.4.6) since every independence statement in the BN is true for the corresponding graph. However, it is not a D-MAP since $t_1 \perp\!\!\!\perp t_2 | y_2$ cannot be inferred from the BN. Similarly no undirected graph can represent all independence statements true in (4.4.6). In this case no perfect MAP (a BN or a MN) can represent (4.4.6).

4.5 Factor Graphs

Factor Graphs (FGs) are mainly used as part of inference algorithms².

Definition 35 (*Factor Graph*). Given a function

$$f(x_1, \dots, x_n) = \prod_i \psi_i(\mathcal{X}^i) \quad (4.5.1)$$

The FG has a node (represented by a square) for each factor ψ_i , and a variable node (represented by a circle) for each variable x_j . For each $x_j \in \mathcal{X}^i$ an undirected link is made between factor ψ_i and variable x_j .

For a factor $\psi_i(\mathcal{X}^i)$ which is a conditional distribution $p(x_i | \text{pa}(x_i))$, we may use directed links from the parents to the factor node, and a directed link from the factor node to the child. This has the same structure as an (undirected) FG, but preserves the information that the factors are distributions.

Factor Graphs are useful since they can preserve more information about the form of the distribution than either a Belief Network or a Markov Network (or Chain Graph) can do alone.

Consider the distribution

$$p(a, b, c) = \phi(a, b)\phi(a, c)\phi(b, c) \quad (4.5.2)$$

²Formally a FG is an alternative graphical depiction of a hypergraph[81] in which the vertices represent variables, and a hyperedge a factor as a function of the variables associated with the hyperedge. A FG is therefore a hypergraph with the additional interpretation that the graph represents a function defined as products over the associated hyperedges. Many thanks to Robert Cowell for this observation.

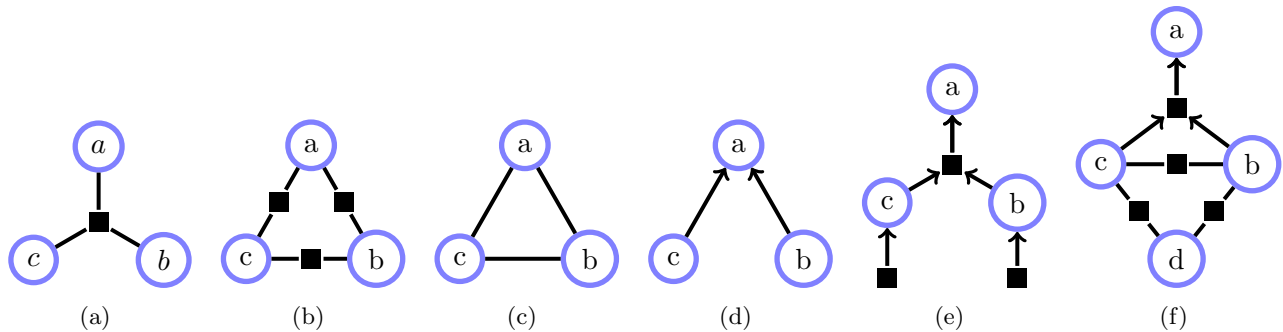


Figure 4.9: **(a)**: $\phi(a, b, c)$. **(b)**: $\phi(a, b)\phi(b, c)\phi(c, a)$. **(c)**: $\phi(a, b, c)$. Both (a) and (b) have the same undirected graphical model, (c). **(e)**: Directed FG of the BN in (d). (a) is an undirected FG of (d). The advantage of (e) over (a) is that information regarding the marginal independence of variables b and c is clear from graph (e), whereas one could only ascertain this by examination of the numerical entries of the factors in graph (a). **(f)**: A partially directed FG of $p(a|b, c)\phi(d, c)\phi(b, d)$. No directed, undirected or chain graph can represent both the conditional and marginal independence statements expressed by this graph and also the factored structure of the undirected terms.

The MN representation is given in fig(4.9c). However, fig(4.9c) could equally represent some unfactored clique potential $\phi(a, b, c)$. In this sense, the FG representation in fig(4.9b) more precisely conveys the form of distribution equation (4.5.2). An unfactored clique potential $\phi(a, b, c)$ is represented by the FG fig(4.9a). Hence different FGs can have the same MN since information regarding the structure of the clique potential is lost in the MN.

4.5.1 Conditional independence in factor graphs

A rule which works with both directed and undirected (and partially directed) FGs is as follows[96]. To determine whether two variables are independent given a set of conditioned variables, consider all paths connecting the two variables. If all paths are blocked, the variables are conditionally independent.

A path is blocked if any one or more of the following conditions are satisfied:

- One of the variables in the path is in the conditioning set.
- One of the variables or factors in the path has two incoming edges that are part of the path, and neither the variable or factor nor any of its descendants are in the conditioning set.

4.6 Notes

A detailed discussion of the axiomatic and logical basis of conditional independence is given in [45] and [264].

4.7 Code

`condindep.m`: Conditional Independence test $p(X, Y|Z) = p(X|Z)p(Y|Z)$?

4.8 Exercises

Exercise 37. 1. Consider the pairwise Markov Network,

$$p(x) = \phi(x_1, x_2)\phi(x_2, x_3)\phi(x_3, x_4)\phi(x_4, x_1) \quad (4.8.1)$$

Express in terms of ϕ the following:

$$p(x_1|x_2, x_4), \quad p(x_2|x_1, x_3), \quad p(x_3|x_2, x_4), \quad p(x_4|x_1, x_3) \quad (4.8.2)$$

2. For a set of local distributions defined as

$$p_1(x_1|x_2, x_4), \quad p_2(x_2|x_1, x_3), \quad p_3(x_3|x_2, x_4), \quad p_4(x_4|x_1, x_3) \quad (4.8.3)$$

is it always possible to find a joint distribution $p(x_1, x_2, x_3, x_4)$ consistent with these local conditional distributions?

Exercise 38. Consider the Markov network

$$p(a, b, c) = \phi_{ab}(a, b)\phi_{bc}(b, c) \quad (4.8.4)$$

Nominally, by summing over b , the variables a and c are dependent. For binary b , explain a situation in which this is not the case, so that marginally, a and c are independent.

Exercise 39. Show that for the Boltzmann machine

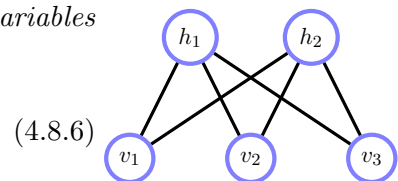
$$p(\mathbf{x}) = \frac{1}{Z(\mathbf{W}, \mathbf{b})} e^{\mathbf{x}^T \mathbf{W} \mathbf{x} + \mathbf{x}^T \mathbf{b}} \quad (4.8.5)$$

one may assume, without loss of generality, $\mathbf{W} = \mathbf{W}^T$.

Exercise 40.

The **restricted Boltzmann machine** (or **Harmonium**[253]) is a specially constrained Boltzmann machine on a bipartite graph, consisting of a layer of visible variables $\mathbf{v} = (v_1, \dots, v_V)$ and hidden variables $\mathbf{h} = (h_1, \dots, h_H)$:

$$p(\mathbf{v}, \mathbf{h}) = \frac{1}{Z(\mathbf{W}, \mathbf{a}, \mathbf{b})} e^{\mathbf{v}^T \mathbf{W} \mathbf{h} + \mathbf{a}^T \mathbf{v} + \mathbf{b}^T \mathbf{h}} \quad (4.8.6)$$



All variables are binary taking states 0, 1.

1. Show that the distribution of hidden units conditional on the visible units factorises as

$$p(\mathbf{h}|\mathbf{v}) = \prod_i p(h_i|\mathbf{v}), \quad \text{with } p(h_i|\mathbf{v}) = \sigma \left(b_i + \sum_j W_{ji} v_j \right) \quad (4.8.7)$$

where $\sigma(x) = e^x / (1 + e^x)$.

2. By symmetry arguments, write down the form of the conditional $p(\mathbf{v}|\mathbf{h})$.

3. Is $p(\mathbf{h})$ factorised?

4. Can the partition function $Z(\mathbf{W}, \mathbf{a}, \mathbf{b})$ be computed efficiently for the RBM?

Exercise 41. Consider

$$p(x) = \phi(x_1, x_{100}) \prod_{i=1}^{99} \phi(x_i, x_{i+1}) \quad (4.8.8)$$

Is it possible to compute $\operatorname{argmax}_{x_1, \dots, x_{100}} p(x)$ efficiently?

Exercise 42. You are given that

$$x \perp\!\!\!\perp y | (z, u), \quad u \perp\!\!\!\perp z | \emptyset \quad (4.8.9)$$

Derive the most general form of probability distribution $p(x, y, z, u)$ consistent with these statements. Does this distribution have a simple graphical model?

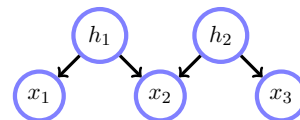
Exercise 43. The undirected graph  represents a Markov Network with nodes x_1, x_2, x_3, x_4, x_5 , counting clockwise around the pentagon with potentials $\phi(x_i, x_{1+\text{mod}(i,5)})$. Show that the joint distribution can be written as

$$p(x_1, x_2, x_3, x_4, x_5) = \frac{p(x_1, x_2, x_5)p(x_2, x_4, x_5)p(x_2, x_3, x_4)}{p(x_2, x_5)p(x_2, x_4)} \quad (4.8.10)$$

and express the marginal probability tables explicitly as functions of the potentials $\phi(x_i, x_j)$.

Exercise 44.

Consider the Belief Network on the right.



1. Write down a Markov Network of $p(x_1, x_2, x_3)$.
2. Is your Markov Network a perfect map of $p(x_1, x_2, x_3)$?

Exercise 45. Two research labs work independently on the relationship between discrete variables x and y . Lab A proudly announces that they have ascertained distribution $p_A(x|y)$ from data. Lab B proudly announces that they have ascertained $p_B(y|x)$ from data.

1. Is it always possible to find a joint distribution $p(x, y)$ consistent with the results of both labs?
2. Is it possible to define consistent marginals $p(x)$ and $p(y)$, in the sense that $p(x) = \sum_y p_A(x|y)p(y)$ and $p(y) = \sum_x p_B(y|x)p(x)$? If so, explain how to find such marginals. If not, explain why not.

Exercise 46. Research lab A states its findings about a set of variables $x_1, \dots, x_n$ as a list L_A of conditional independence statements. Lab B similarly provides a list of conditional independence statements L_B .

1. Is it possible to find a distribution which is consistent with L_A and L_B ?
2. If the lists also contain dependence statements, how could one attempt to find a distribution that is consistent with both lists?

Exercise 47.

Consider the distribution

$$p(x, y, w, z) = p(z|w)p(w|x, y)p(x)p(y) \quad (4.8.11)$$

1. Write $p(x|z)$ using a formula involving (all or some of) $p(z|w)$, $p(w|x, y)$, $p(x)$, $p(y)$.
2. Write $p(y|z)$ using a formula involving (all or some of) $p(z|w)$, $p(w|x, y)$, $p(x)$, $p(y)$.
3. Using the above results, derive an explicit condition for $x \perp\!\!\!\perp y | z$ and explain if this is satisfied for this distribution.

Exercise 48. Consider the distribution

$$p(t_1, t_2, y_1, y_2, h) = p(y_1|t_1, h)p(y_2|t_2, h)p(t_1)p(t_2)p(h) \quad (4.8.12)$$

1. Draw a Belief Network for this distribution.
2. Can the distribution

$$p(t_1, t_2, y_1, y_2) = \sum_h p(y_1|t_1, h)p(y_2|t_2, h)p(t_1)p(t_2)p(h) \quad (4.8.13)$$

be written as a ('non-complete') Belief Network?

3. Show that for $p(t_1, t_2, y_1, y_2)$ as defined above $t_1 \perp\!\!\!\perp y_2 | \emptyset$.

Exercise 49. Consider the distribution

$$p(a, b, c, d) = \phi_{ab}(a, b)\phi_{bc}(b, c)\phi_{cd}(c, d)\phi_{da}(d, a) \quad (4.8.14)$$

where the ϕ are potentials.

1. Draw a Markov Network for this distribution.
2. Explain if the distribution can be represented as a ('non-complete') Belief Network.
3. Derive explicitly if $a \perp\!\!\!\perp c \mid \emptyset$.

Exercise 50. Show how for any singly-connected Markov network, one may construct a Markov equivalent Belief Network.

Exercise 51. Consider a pairwise binary Markov network defined on variables $s_i \in \{0, 1\}$, $i = 1, \dots, N$, with $p(s) = \prod_{ij \in \mathcal{E}} \phi_{ij}(s_i, s_j)$, where $\mathcal{E}$ is a given edge set and the potentials ϕ_{ij} are arbitrary. Explain how to translate such a Markov network into a Boltzmann machine.